

Literature-Implied Answer for arXiv:math/0502081

run fa_banach_001, agent agent_lane_06

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Status

Status: literature-implied answer, negative for the no- ℓ_1 $\mathcal{L}_\infty$ /non- $\mathcal{LP}$ subquestion.

Source Question

In Section 4 of Castillo–Moreno–Suarez, *On Lindenstrauss–Pelczynski spaces*, arXiv:math/0502081, after the corollary that twisted sums of $C(K)$ -spaces are $\mathcal{LP}$ -spaces, the authors state:

“We do not know if there exist $\mathcal{L}_\infty$ -spaces not containing ℓ_1 which are not $\mathcal{LP}$ spaces.”

In the parsed source this is the paragraph around lines 817–819 of `data/parsed/arxiv_sources/0502081/source`.

Supporting Literature

Castillo–Papini, *On isomorphically polyhedral $\mathcal{L}_\infty$ -spaces*, arXiv:1601.02035, proves in its final section:

There is an isomorphically polyhedral $\mathcal{L}_\infty$ -space that is not an $\mathcal{LP}$ -space.

The proof constructs, for a subspace $H \subset c_0$, an exact sequence

$$0 \longrightarrow C(\omega^\omega) \longrightarrow \Omega_H \longrightarrow c_0 \longrightarrow 0$$

and shows that Ω_H is not an $\mathcal{LP}$ -space.

Identification

The constructed space Ω_H is an $\mathcal{L}_\infty$ -space and is not $\mathcal{LP}$ by arXiv:1601.02035. It remains only to check the “not containing ℓ_1 ” clause from the source question. The kernel $C(\omega^\omega)$ contains no copy of ℓ_1 : its compact space is countable and scattered, so its dual is separable, whereas any Banach space containing ℓ_1 has nonseparable dual. The quotient c_0 contains no copy of ℓ_1 . Finally, containing no copy of ℓ_1 is a three-space property: if $0 \rightarrow Y \rightarrow X \rightarrow Z \rightarrow 0$ has Y and Z with no ℓ_1 , then an ℓ_1 -copy in X either maps isomorphically into Z , or a block basis is mapped arbitrarily close to Y , yielding an ℓ_1 -copy in Y .

Thus Ω_H is an $\mathcal{L}_\infty$ -space containing no ℓ_1 , but it is not a Lindenstrauss–Pelczynski space. This gives a negative answer to the source subquestion.

Scope and Limitations

This is not a solution of the later quotient questions in Section 5 of arXiv:math/0502081. In particular, it does not decide whether quotients of two $\mathcal{LP}$ -spaces are $\mathcal{LP}$, whether $\ell_\infty/C[0,1]$ is $\mathcal{LP}$, or whether the $c_0(\Gamma)$ -extension property holds for all $\mathcal{LP}$ ranges.

The relation is an agent-identified implication. The supporting paper cites the source paper and proves the needed example, but does not explicitly state that it is resolving this particular sentence from arXiv:math/0502081.

References

1. J. M. F. Castillo, Y. Moreno, J. Suarez, *On Lindenstrauss-Pelczynski spaces*, arXiv:math/0502081.
2. J. M. F. Castillo and P. L. Papini, *On isomorphically polyhedral $\mathcal{L}_\infty$ -spaces*, arXiv:1601.02035.